How to use: spotiflow_batch.py (optional)

Reproducible Spotiflow detection for the whole project: one calibrated setting per camera setup instead of thresholds chosen per image. pipeline version 4 (October 2026).

Optional. The current analysis uses your own napari/Spotiflow detections. Use this script if you redetect the project. **Output feeds:** Fish_spots_batch_v4.py.

Status: the Spotiflow call itself could not be tested in the development environment (no PyTorch); everything around it was tested with a stand-in detector. On a first real run, check the calibration figures carefully, and report any error in the predict call. A first trial on the full data set produced overdetection; treat the recommended settings as a starting point and validate them visually.

1. Requirements

Run in the Python environment where napari-spotiflow works (Spotiflow + PyTorch). CPU is fine (overnight for ~1,300 images). The script sets KMP_DUPLICATE_LIB_OK to avoid the Windows "OMP: Error #15".

2. Three modes, in order

Command	Time	Does
`python spotiflow_batch.py inventory`	seconds	lists images with soma mask, groups them by camera setup and control/protease pair, warns about anomalies
`python spotiflow_batch.py calibrate`	hours (resumable)	random calibration images (2 per setup × probe × condition + 1 per pair not yet sampled); 12 settings (model, 2× downsampling, normalisation); recall of bright reference spots; writes recommended_settings.json
`python spotiflow_batch.py detect`	overnight (resumable)	detects every image with the recommended settings; old spot CSVs moved to _old_spots (never deleted)

3. Checking the calibration (folder _spotiflow_calibration)

1. 1_reference_intensity_populations.png - red cutoff between dim and bright population for every setup × probe (one line per pair). Wrong? edit cutoff_used in reference_cutoffs.csv, rerun calibrate (minutes, cached).
2. 2_recall_vs_threshold_<setup>.png - recall ≥ 95% at the chosen threshold; third panel = fraction of detections that are dim puncta.
3. 3_detections_<setup>.png - zoom per calibration image: green found, red missed real spots.
4. Edit recommended_settings.json if needed; detect uses whatever is in it.

Setting	Meaning
MODELS, DOWNSAMPLE_OPTIONS, NORM_H_OPTIONS	calibration grid
MIN_RECALL / MEDIAN_RECALL	0.95 / 0.98 - required bright-spot recall
CAL_IMAGES_PER_PROBE_CONDITION, CAL_ENSURE_EVERY_PAIR	calibration sampling
DEVICE	'cpu', 'cuda' or 'mps'

To go back to your own detections: restore_original_spots.py (helper-scripts guide).